

Algorithmic Bias in Educational Technology and Student Surveillance

The Issue

AI-powered learning platforms, remote proctoring software, and predictive analytics systems are now deeply embedded in educational institutions, spanning from K-12 to higher education. While they are often marketed as objective, efficient, and innovative solutions to longstanding educational challenges, these technologies can unintentionally magnify existing inequalities, perpetuate systemic discrimination, and infringe on student privacy in novel and troubling ways. The so-called "black box" nature of many algorithms means that bias is not only present but also invisible and unaccountable. Students, parents, and even educators often have little understanding of—or control over—how vast amounts of personal and academic data are collected, processed, and used to make critical decisions. These decisions, which may include admissions, academic placements, and disciplinary actions, can profoundly shape students' educational experiences and long-term opportunities. Moreover, the increasing reliance on algorithmic systems risks shifting power away from human judgment and eroding trust between students and educational institutions.

Why It Matters

Algorithmic Bias: The Illusion of Objectivity

- Algorithms used for college admissions, course recommendations, or "at-risk" student identification can perpetuate racial and socioeconomic bias. While presented as neutral and data-driven, these systems are trained on historical data that reflects existing inequalities—meaning they learn to replicate discriminatory patterns.
- Admissions algorithms may disadvantage applicants from underrepresented backgrounds. When AI systems evaluate applications, they may weigh factors like high school prestige, zip code, extracurricular activities, or writing style in ways that systematically favor wealthy, white applicants. Students from under-resourced schools or neighborhoods get algorithmically sorted out before human eyes ever see their applications.
- "At-risk" student identification systems can create self-fulfilling prophecies. Predictive analytics flag students as likely to fail or drop out based on factors like demographics, prior test scores, attendance, or engagement metrics. Once flagged, students may receive interventions that are stigmatizing, lower expectations from teachers, or reduced access to advanced courses—reinforcing the very outcomes the algorithm predicted.
- Course recommendation algorithms track students into paths that limit opportunity. AI systems may steer students toward remedial courses, vocational tracks, or "easier" options based on biased assessments of their potential, perpetuating educational stratification along race and class lines.
- Grading and assessment algorithms may penalize diverse communication styles. Automated essay scoring systems often favor formulaic, standardized academic English, potentially marking down students who use African American Vernacular English (AAVE), code-switching, creative expression, or non-Western rhetorical structures—even when ideas are strong and clear.
- AI tutoring and adaptive learning platforms may provide lower-quality instruction to certain students. If algorithms determine students are "struggling," they may be routed to simpler content, repetitive drill-and-practice exercises, or slower pacing—denying them access to challenging, enriching material that could help them grow.
- The bias is often invisible and unaccountable. Proprietary algorithms are treated as trade secrets, so schools, teachers, students, and families cannot examine how decisions are made, challenge biased outcomes, or demand transparency.

Remote Proctoring: Surveillance Disguised as Academic Integrity

- Remote proctoring software disproportionately flags students of color and monitors students in invasive ways (scanning rooms, tracking eye movements). These systems use facial recognition, eye-tracking, audio monitoring, and behavioral analysis to detect "cheating," but the technology is rife with problems.
- Facial recognition technology fails more often for people with darker skin tones, a well-documented problem across AI systems. Black, Latino, Asian, and Indigenous students experience higher rates of false flags, technical failures, or inability to log in—causing exam anxiety, delays, accusations, or outright exam failure through no fault of their own. (Chhua et al., 2024)
- Students must show their faces on camera for extended periods, which may be uncomfortable or impossible for students who wear religious head coverings, have facial differences or disabilities, or experience gender dysphoria.
- Eye-tracking algorithms flag "suspicious" behavior like looking away from the screen, but students with ADHD, autism, anxiety, or visual processing differences may naturally move their eyes differently. Neurotypical assumptions are coded into what counts as "normal" test-taking behavior.
- Room scans invade students' private spaces. Proctoring software often requires students to pan their webcam around the room to prove no one else is present and no unauthorized materials are visible. This forces students to reveal their living conditions—small apartments, shared bedrooms, family members in the background, lack of furniture, visible poverty—to a remote proctor and potentially recorded permanently.
- Background noise and interruptions are flagged as suspicious. Students living in multigenerational households, with roommates, in noisy neighborhoods, or with young siblings may have their exams interrupted or flagged for sounds beyond their control.
- Low-income students may lack private, quiet spaces for monitored exams, further disadvantaging them. A student taking an exam from a car, a library parking lot, or a crowded kitchen is already stressed and may be penalized by proctoring algorithms that expect ideal testing environments.
- Invasive monitoring includes keystroke analysis, browser lockdowns, and audio recording. Some systems prevent students from accessing any other applications, read clipboard data, or continuously record audio "just in case." Students have been flagged for using calculators on their phone (when permitted), talking to themselves (a common study technique), or standing up to stretch.
- Students with disabilities face particular barriers. Those who need screen readers, alternative input devices, movement breaks, or testing accommodations may find proctoring software incompatible with their needs or may be flagged for "unusual" behavior that is actually their accommodation.
- The psychological toll is significant. Being watched continuously creates anxiety that impairs performance. Students report feeling dehumanized, violated, and distrusted. Some experience panic attacks during proctored exams. (ACLU RESEARCH REPORT, n.d.)
- False accusations of cheating have serious consequences: failing the exam, failing the course, academic integrity violations on permanent records, suspension, or expulsion. Appealing these decisions is often difficult, especially when the "evidence" is algorithmic scores rather than clear proof.

Data Collection and Privacy Violations

- Data collected on students—from browsing history to biometric data—raises privacy concerns and can be sold to third parties. Educational technology companies gather massive amounts of data on students, often with minimal oversight or informed consent.
- The scope of data collection is staggering: names, birthdates, addresses, biometric identifiers (face scans, voice prints, keystroke patterns), academic records, behavioral data (time on task, clicks, mistakes), social-emotional data (mood

check-ins, survey responses), health information, disciplinary records, communications, location data, and more. (Nhando, 2026)

- Students may not understand how their data is used or have control over it. Privacy policies are long, legalistic, and rarely read. Students (and parents) may not realize they're consenting to extensive surveillance and data mining when they click "agree" to use a learning platform.
- Third-party vendors have inconsistent privacy protections. Educational data may be stored on servers with weak security, shared with advertisers, sold to data brokers, or used to build behavioral profiles that follow students beyond school.
- FERPA (Family Educational Rights and Privacy Act) protections have loopholes. While FERPA regulates educational records, it has broad "school official" exceptions that allow third-party companies to access student data if they provide services to schools. Once data leaves the school's control, protections weaken. (FERPA exceptions, n.d.)
- Student data can be breached or hacked. Educational technology companies are frequent targets of cyberattacks. Breaches can expose sensitive information about minors, including their locations, routines, and vulnerabilities. (FTC Takes Action Against Education Technology Provider for Failing to Secure Students' Personal Data, 2025)
- Data persistence is a concern. Information collected in elementary school may follow students through college and into employment. Behavioral flags, "risk scores," or disciplinary data could haunt students for years, affecting opportunities in ways they never know about.
- Predictive profiles can limit futures. Data analytics companies build psychological and behavioral profiles of students that may be used by colleges, employers, or others to make decisions—turning childhood mistakes or struggles into permanent digital scarlet letters.

Surveillance Culture and Its Impact on Learning

- Constant monitoring changes student behavior in harmful ways. When students know they're being watched and assessed at all times, they may become risk-averse, less creative, and less willing to experiment or make mistakes—all essential to learning.
- Trust between students and institutions erodes. Surveillance communicates that students are presumed cheaters, threats, or problems to be managed rather than learners to be supported. This undermines the educational relationship.
- Surveillance disproportionately impacts already-marginalized students. Black and Brown students, students with disabilities, LGBTQ+ students, and low-income students are most likely to be surveilled, flagged, and punished. This reinforces school-to-prison pipeline dynamics and systemic oppression.
- Students internalize surveillance. Growing up under constant digital monitoring normalizes invasive tracking, teaching students that privacy is not a right and that they should accept surveillance as a condition of participation in society.
- Critical thinking and dissent are chilled. Students may self-censor in online discussions, avoid controversial topics, or hesitate to challenge ideas if they know their participation is tracked, analyzed, and potentially used against them.

The School-to-Prison Pipeline Goes Digital

- Behavioral monitoring algorithms can flag students for "concerning" online activity, such as researching mental health, LGBTQ+ identity, protest movements, or even assigned coursework on topics like terrorism or revolution. These flags may be reported to school administrators, parents, or law enforcement. (Schools use AI to monitor kids, hoping to prevent violence. Our investigation found security risks, 2025)
- Students of color are disproportionately flagged for the same behaviors that white students engage in without consequence, mirroring disparities in school discipline more broadly. (Brislin et al., 2024, pp. 1225-1238)

- Law enforcement partnerships with ed-tech companies create pathways from classroom to criminal justice system. Some schools use software that directly notifies police of flagged behavior, turning educational surveillance into criminal surveillance.
- Students have been visited by police, interrogated, or even arrested based on algorithmic flags—sometimes for creative writing assignments, song lyrics, or venting frustration online.

Economic and Educational Inequities

- Wealthy schools and students benefit most from educational technology while bearing fewer burdens. They access premium platforms, have IT support, and can opt out of invasive tools by switching to private schools or hiring tutors.
- Low-income schools are more likely to rely on free or low-cost ed-tech that monetizes student data as part of its business model. Students in under-resourced schools thus face both more surveillance and worse technology. (Prothero, 2024)
- The digital divide intersects with surveillance. Students with limited internet or devices may struggle to comply with proctoring requirements, while also facing algorithmic penalties for inconsistent engagement with online platforms.
- "Personalized learning" can become "personalized inequality." Adaptive algorithms that sort students into differentiated learning paths may entrench rather than address achievement gaps, giving some students rigorous challenges while others get watered-down content.

Lack of Regulation and Accountability

- Educational technology is largely unregulated, with regulatory frameworks lagging far behind the rapid pace of technological change. Unlike medical devices or financial services, ed-tech faces minimal oversight regarding efficacy, safety, bias, or privacy. Companies can deploy experimental, unproven algorithms on students without demonstrating that these systems are effective, fair, or safe. There are few requirements for independent validation or public reporting of outcomes.
- Schools often lack the expertise and resources to critically evaluate ed-tech claims. Administrators, teachers, and school boards may not have the technical knowledge or time to assess the presence of algorithmic bias, data security vulnerabilities, or privacy implications. This lack of capacity leaves schools reliant on vendor marketing materials and sales pitches, which may overstate benefits and minimize risks.
- Procurement processes frequently prioritize cost, convenience, and speed over student welfare. Cash-strapped districts may adopt ed-tech tools simply because they are cheap or free, sometimes as a condition for access to basic digital infrastructure. These decisions are often made without thorough vetting of privacy policies, ongoing maintenance needs, or consideration of long-term ethical implications. Vendors may bundle products or offer incentives that make it difficult for schools to opt out of problematic systems.
- Students and families have little recourse when algorithms make mistakes, reinforce bias, or violate privacy. Challenging these decisions is often difficult or impossible. Companies may hide behind proprietary claims, preventing external audits or appeals. Schools themselves may lack clear mechanisms for students and families to contest algorithmic decisions, seek remedies, or have data deleted.
- Audit and transparency requirements remain minimal or nonexistent. Few jurisdictions mandate algorithmic impact assessments, independent bias audits, or public disclosure of how ed-tech systems make decisions and what data they use. (Algorithmic Accountability Act of 2025, n.d.) As a result, potentially harmful systems can operate unchecked, and meaningful accountability is rare.

Resistance and Harm to Marginalized Communities

- Indigenous students' data sovereignty is violated. Tribal nations have specific concerns about data collection on Indigenous children, which may be used without consent or cultural understanding, and which violates principles of Indigenous data sovereignty.
- Undocumented students and mixed-status families face unique risks. Data collection that includes addresses, family information, or immigration status could be shared with immigration enforcement, deliberately or through breaches.
- Students with disabilities face both exclusion and over-surveillance. Inaccessible platforms exclude them from learning, while special education data is often shared extensively among staff and systems, violating privacy.
- LGBTQ+ students risk being outed. Algorithms that analyze writing, social patterns, or search history could identify students exploring their identity before they're ready to disclose, potentially putting them at risk in unsupportive homes or communities.

The Pedagogy of Surveillance vs. The Pedagogy of Trust

- Surveillance-based education assumes students are problems to manage rather than humans to nurture. It prioritizes compliance, control, and risk mitigation over curiosity, growth, and agency.
- Trust-based education values relationships, autonomy, and intrinsic motivation. It recognizes that learning requires safety to take risks, make mistakes, and explore ideas without fear of permanent consequences.
- The presence of surveillance changes what education means. When every click, glance, and word is monitored and analyzed, education becomes performance for algorithms rather than authentic intellectual and personal development.

What Students and Families Deserve

- **Transparency**
 - Clear, accessible information about what data is collected on students.
 - Explanation of how data is used, who has access, and how long it is retained.
 - Regular reporting to students and families about data practices.
- **Consent**
 - Meaningful opt-in (not just opt-out) for non-essential data collection and surveillance.
 - Informed consent processes that ensure students and families truly understand what they're agreeing to.
 - The right to withdraw consent at any time without penalty.
- **Algorithmic Accountability**
 - Independent audits of algorithms for bias, accuracy, and fairness.
 - Public reporting of outcomes by demographic group to identify disparities.
 - Transparent explanations of how algorithmic decisions are made.
- **Data Minimization**
 - Only collecting data that is truly necessary for educational purposes.
 - Prohibiting the use of student data for marketing or non-educational purposes.
 - Regular deletion of outdated or unnecessary data.
- **Privacy by Design**
 - Ed-tech tools should be built with strong privacy protections as the default, not as an afterthought.

- End-to-end encryption and secure data storage.
- Minimizing the number of people and vendors who can access sensitive information.
- Right to Access and Deletion
 - Students and families should be able to see all data collected about them in a clear, accessible format.
 - The right to request correction or deletion of inaccurate or unwanted data.
 - Mechanisms for appealing or contesting data-driven decisions.
- Human Decision-Making
 - Important decisions (admissions, placement, discipline) should not be made by algorithms alone.
 - Human oversight and robust appeal processes must be in place.
 - Students and families should be able to speak directly with decision-makers.
- Equity Impact Assessments
 - Before adopting new ed-tech, schools should assess potential harms to marginalized students.
 - Ongoing monitoring for disparate impacts on race, class, disability, gender, and immigration status.
 - Involving students and families from diverse backgrounds in decision-making.
- Alternatives to Invasive Proctoring
 - Implementing honor codes, open-note exams, and alternative assessments.
 - Designing assessments that reduce the need for surveillance.
 - Rebuilding cultures of academic integrity based on trust, not suspicion.
- Regulation and Enforcement
 - Strong laws protecting student privacy and prohibiting discriminatory algorithms.
 - Real penalties for violations, including fines and loss of contracts.
 - Independent oversight bodies to investigate complaints and enforce standards.

The Bottom Line

Educational technology is not neutral. Algorithms embed the biases of their creators and the data they're trained on, and surveillance systems treat students as subjects to monitor rather than people to empower. The rush to digitize education has outpaced ethical consideration, regulatory protection, and evidence of benefit. Students—especially those already marginalized by race, class, disability, or immigration status—bear the greatest burdens and risks. As schools increasingly rely on AI and surveillance tools, we must ask: What kind of education do we want? One that tracks, sorts, and controls? Or one that trusts, nurtures, and liberates? The algorithms we deploy on students today are shaping not just their education, but their relationship to power, privacy, and possibility for the rest of their lives.

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